

TAK BRIDGE

Drone GPS to MGRS to TAK | Quick Start | Development build 0.3.0

Reads aircraft GPS over USB using MAVLink, MSP, GHST or CRSF and relays valid, fresh positions as Cursor on Target (CoT) events. The app receives telemetry; only the Direct FC MSP profile sends a fixed GPS read request.

What you need

Item	Requirement
Android device	Android 8.0 or later with USB host/OTG support
Data cable	USB data cable or OTG adapter; charge-only cables cannot carry telemetry
Telemetry source	Compatible controller, transmitter or flight controller with configured GPS output
TAK destination	ATAK/WinTAK/iTAK receiving the configured multicast, or a TAK Server

1. Choose a USB path

USB telemetry source: Connect an already-configured, compatible USB aircraft telemetry output. This guide covers operation of the bridge; device-side feature discovery and enablement are outside its scope. Choose the app profile and protocol that match the stream actually provided by your source.

Direct flight controller: Use MAVLink or passive auto-detect for an already-streaming source. For an MSP source that needs GPS requests, choose **Direct FC - MSP GPS polling**. This polls at 2 Hz. Selecting MSP in a passive profile does not enable polling.

Passive telemetry: Use **Other USB - passive auto-detect** or select the actual protocol for an existing supported stream. RC/stick frames alone cannot provide aircraft GPS.

2. Connect and verify

With the aircraft on the ground and propellers removed, connect the data cable. Press **CONNECT**, select a port if prompted, and approve Android USB access. The picker identifies devices by VID:PID and port. Confirm fresh GPS, fix and satellite status. A heartbeat or byte counter alone does not prove a GPS downlink.

The foreground service keeps an active session running when you switch apps. Use **STOP** or the notification to end it. USB loss stops the session and clears position; reconnect explicitly.

3. Send to TAK

Local multicast: Enable **Local TAK** and configure the client to receive **239.2.3.1:6969**. Network isolation, VPNs or filtering can block delivery. A sent counter proves a socket write; verify the marker in the receiving client.

Server TCP: Enter the server host and configured port (default 8087), then press **TAK**. **Server TLS:** Enable TLS, open **CERTIFICATES**, import the client .p12 with its password and the server CA if needed. Use a host name or IP present in the server certificate. Press **TAK** to unlock/connect. Passwords are memory-only and must be entered again after the process ends.

Reading the screen

Element	Meaning
Coordinate overlay	Tap to cycle MGRS, decimal degrees, DMS and UTM; CoT always uses decimal degrees
Fix / satellites / speed / altitude	Source GPS evidence; MSL altitude stays in display and remarks, HAE remains unknown
Map marker and trail	Aircraft position and heading, with recent breadcrumb history
USB / TAK status and counters	Connection stages and decoded/sent activity; not proof of client display
GPS STALE	No fresh position for more than five seconds; publishing has stopped

Troubleshooting

Problem	Next check
No USB serial interface	Check cable, Android host role and source serial USB mode
Permission denied	Press CONNECT again and approve the Android access request
USB open, no bytes	Verify that the source is providing telemetry; check the selected port and stream settings
Controller channels only	Configure an upstream aircraft GPS stream; stick motion is not GPS
GPS frames, no fix	Check the source GPS fix; invalid or stale positions are not published
TAK marker missing	Verify group/port or server settings, receiving client and network filtering
TLS rejected	Check password, certificate dates, CA trust and host-name match

Practice and Tools

SOFTWARE DEMO exercises telemetry parsing and local CoT previews without USB. It never sends simulated aircraft positions to a network. **DIAGNOSTICS** exports counters without coordinates or credentials. Tools include RF calculators, terrain and mesh planning, generic VTX profiles and a target catalog. Match frequency/power tables and exact upstream target keys to your hardware before using generated configuration.

Physical USB/GPS/TAK acceptance remains pending. See docs/USB_CONTROLLER_SETUP.md and docs/V1_READINESS.md. Compatibility keys and source/license attribution retain their exact upstream spelling.